Supplementary Information

Supplementary Note 1: Extended Overview of the Study

This study introduces MultiDentNet, a unified deep learning framework for automated screening of multiple dental conditions and binary classification of clinically labeled oral lesions (malignant versus Benign). The framework addresses key limitations of manual examinations and single-condition models by leveraging architectural diversity and inter-class dependency modeling to improve diagnostic accuracy and generalizability. MultiDentNet integrates four pre-trained convolutional neural networks (DenseNet121, EfficientNetV2-S, ResNet50, and Inception-V3), enhanced with squeeze-and-excitation (SE) blocks, graph convolutional networks (GCNs) with learnable adjacency matrices, and a multi-task learning strategy. Predictions are combined via a weighted ensemble optimized on validation performance.

The framework was evaluated on a dental dataset comprising 9,439 images across five conditions (caries, gingivitis, hypodontia, tooth discoloration, and ulcers) and a separate oral lesion subset of 940 images, clinically annotated as malignant or non-malignant by oral medicine specialists. Due to its limited size, single-center origin, anatomical scope (lips and tongue only), and absence of universal histopathological confirmation, the oral lesion analysis is presented strictly as a proof-of-concept and should not be interpreted as validated cancer detection.

Performance was assessed using accuracy, Cohen's $\varkappa$, false-negative rate (FNR), interpretability tools (Grad-CAM, t-SNE), and comprehensive ablation studies. MultiDentNet achieved 99.68% accuracy for dental conditions and 98.53% for oral lesion classification, outperforming individual backbones and prior benchmarks. Inter-rater agreement was near-perfect (Cohen's $\varkappa = 1.00$ for dental conditions, 0.97 for oral lesions), with a malignant lesion FNR of 2.4%. Ablation studies confirmed the contribution of SE blocks, GCNs, and multi-task learning. Visualizations highlighted clinically relevant regions, though rare classes such as hypodontia remained more challenging (FNR: 8.57%).

By combining diverse CNN architectures with adaptive dependency modeling, MultiDentNet delivers accurate and interpretable multi-condition screening for dental pathologies. It reduces diagnostic oversights, supports early referral, and accelerates workflows in simulated clinical settings. Future work should emphasize external validation using multi-center, histopathologically confirmed datasets and multimodal integration (e.g., radiographic and clinical metadata).

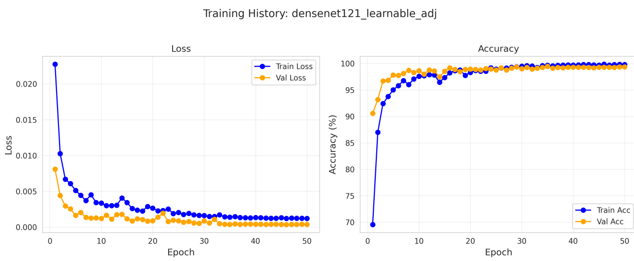
MultiDentNet demonstrates potential as a scalable AI-assisted screening tool with applications in (i) triage in rural or underserved areas, (ii) supporting general practitioners and hygienists in preliminary evaluations, (iii) streamlining high-volume clinic workflows, and (iv) integration into tele-dentistry and mobile health platforms. Its lightweight design enables real-time inference on mobile devices and seamless interfacing with cloud-based electronic health record systems, enhancing continuity of care.

The code and supplementary results are available at `https://github.com/Aliyar4061/MultiDentNet`.

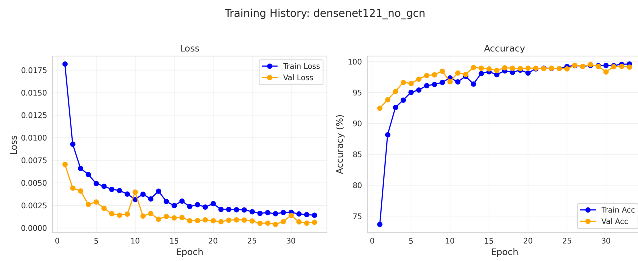
1 Supplementary Figures

1.1 Training Dynamics

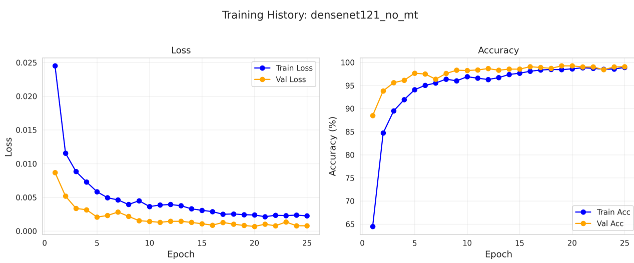
1.1.1 Dental Conditions



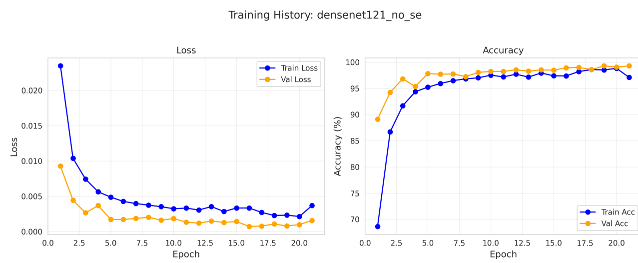
(a) DenseNet121 (Learnable adjacency)



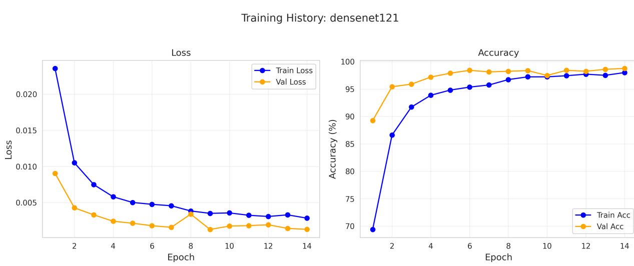
(b) DenseNet121 (No GCN)



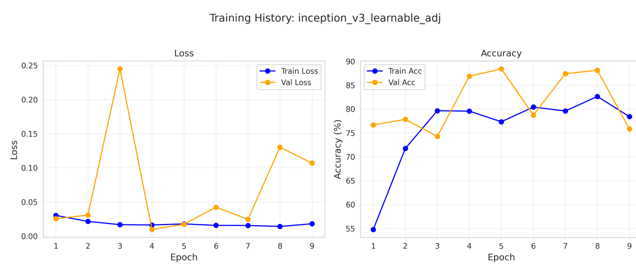
(c) DenseNet121 (No MTL)



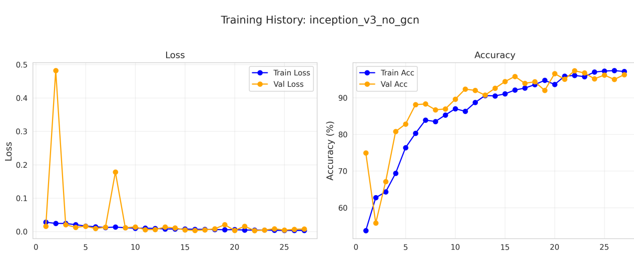
(d) DenseNet121 (No SE)



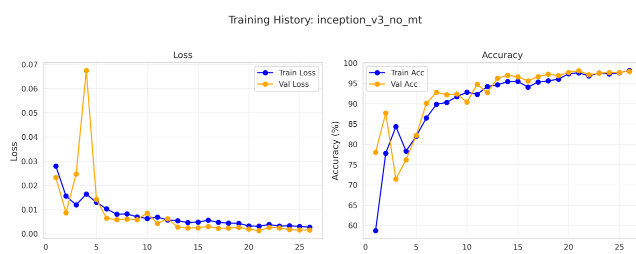
(e) DenseNet121 (Baseline)



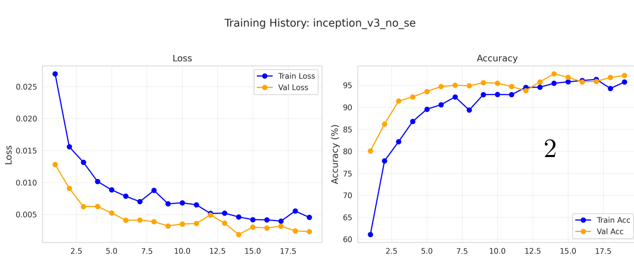
(f) Inception-V3 (Learnable adjacency)



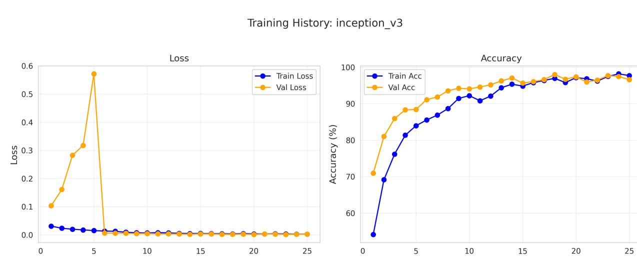
(g) Inception-V3 (No GCN)



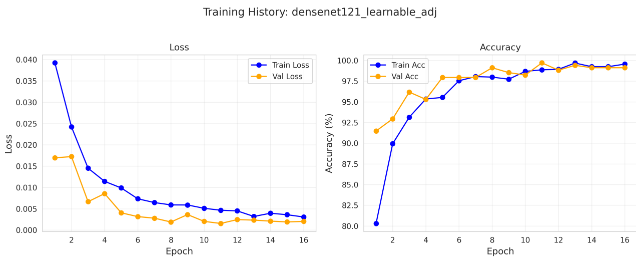
(h) Inception-V3 (No MTL)



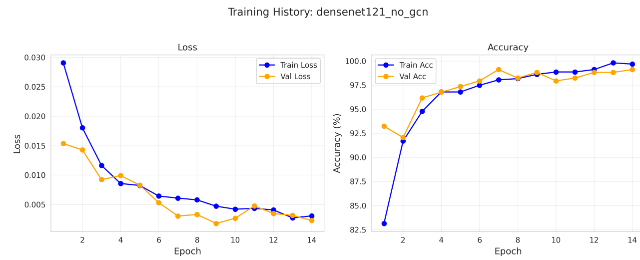
(i) Inception-V3 (No SE)



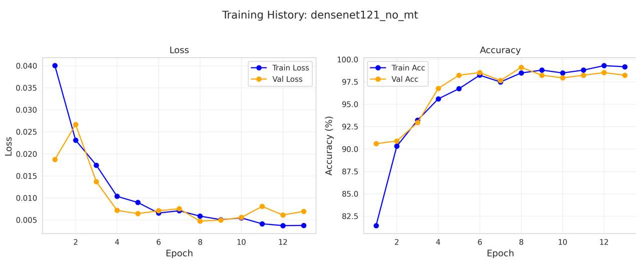
(j) Inception-V3 (Baseline)



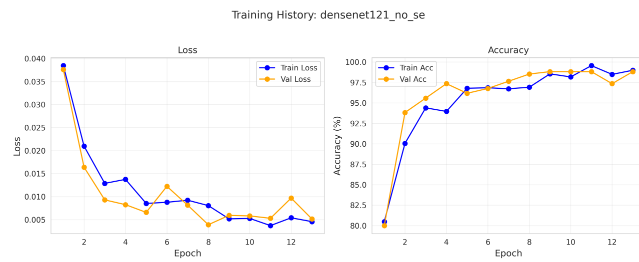
(k) DenseNet121 (Learnable adjacency)



(l) DenseNet121 (No GCN)

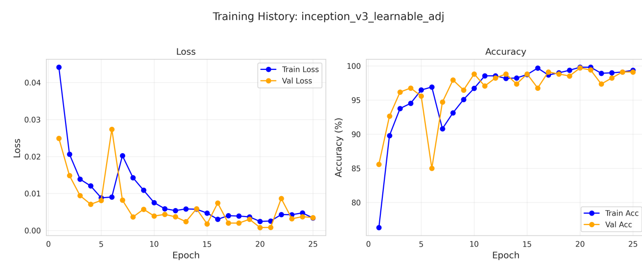


(m) DenseNet121 (No MTL)

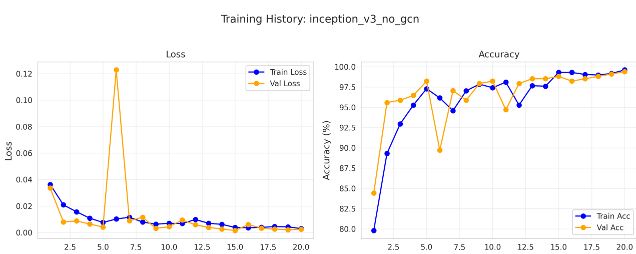


(n) DenseNet121 (No SE)

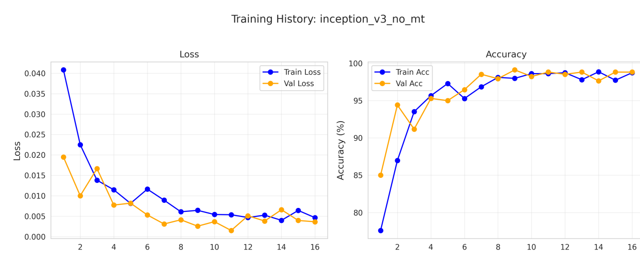
(o) DenseNet121 (Baseline)



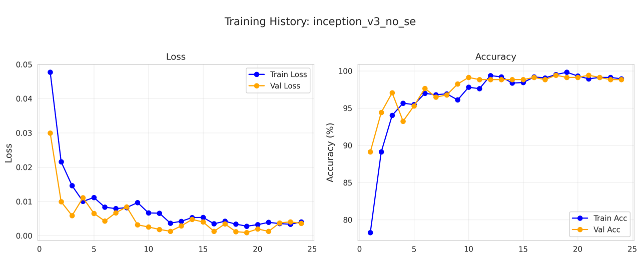
(p) Inception-V3 (Learnable adjacency)



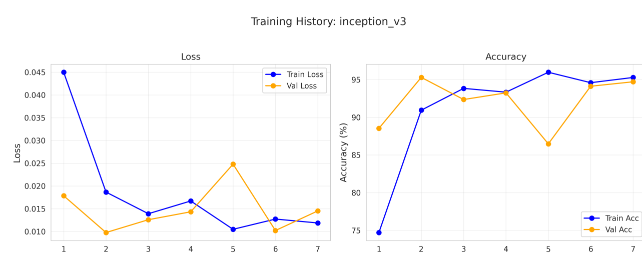
(q) Inception-V3 (No GCN)



(r) Inception-V3 (No MTL)



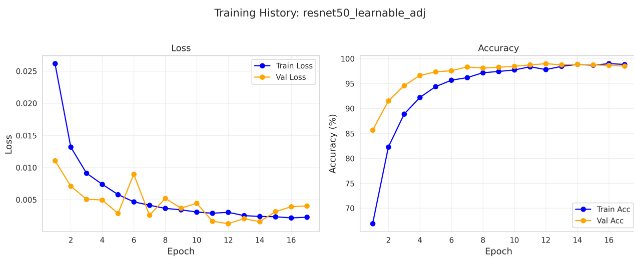
(s) Inception-V3 (No SE)



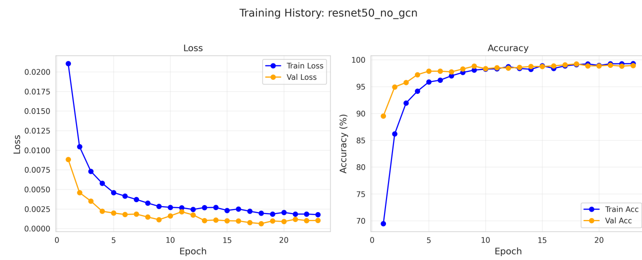
(t) Inception-V3 (Baseline)

Supplementary Fig. S1. Learning curves for dental conditions dataset across model variants

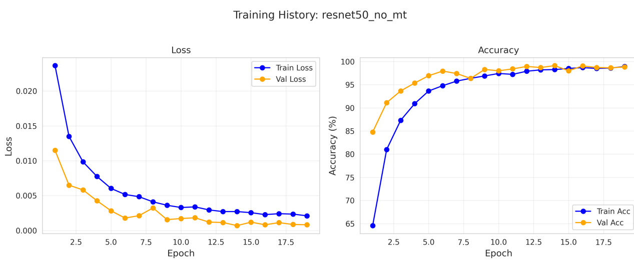
1.1.2 Oral Cancer



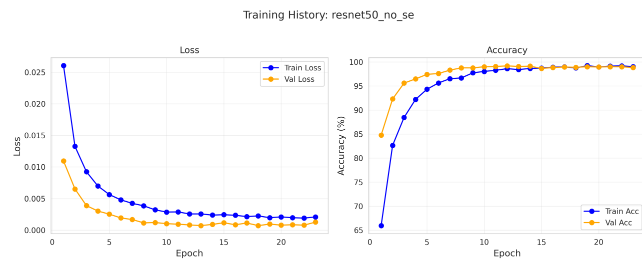
(a) ResNet50 (Learnable adjacency)



(b) ResNet50 (No GCN)



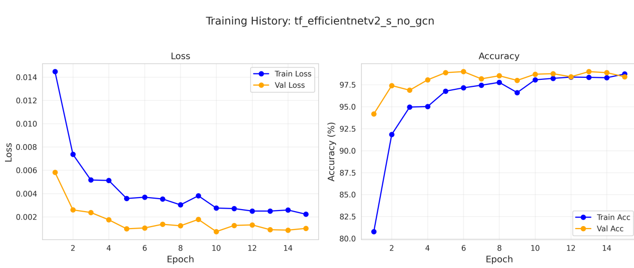
(c) ResNet50 (No MTL)



(d) ResNet50 (No SE)

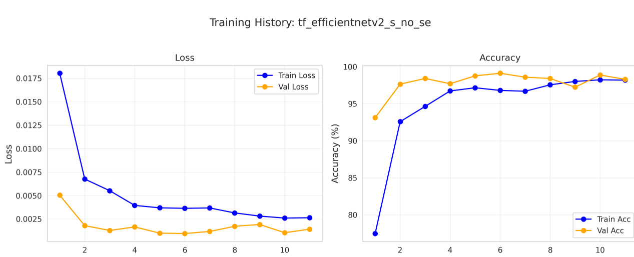
(e) ResNet50 (Baseline)

(f) EfficientNetV2-S (Learnable adjacency)

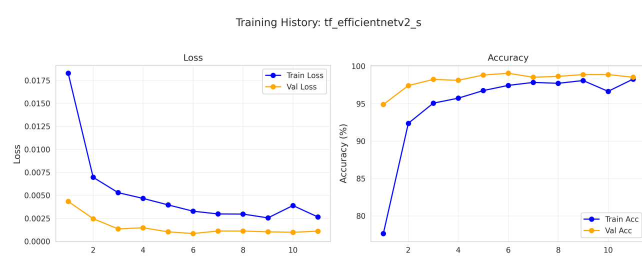


(g) EfficientNetV2-S (No GCN)

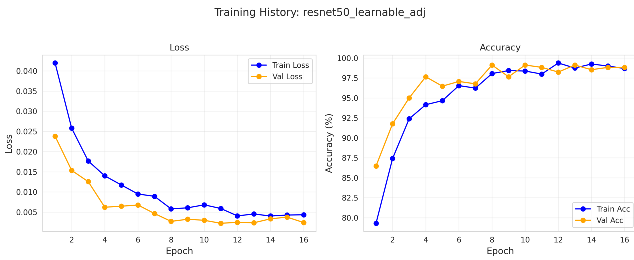
(h) EfficientNetV2-S (No MTL)



(i) EfficientNetV2-S (No SE)

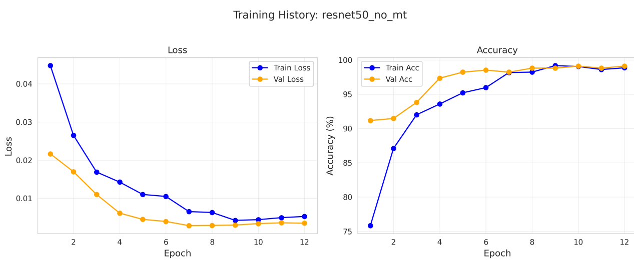


(j) EfficientNetV2-S (Baseline)



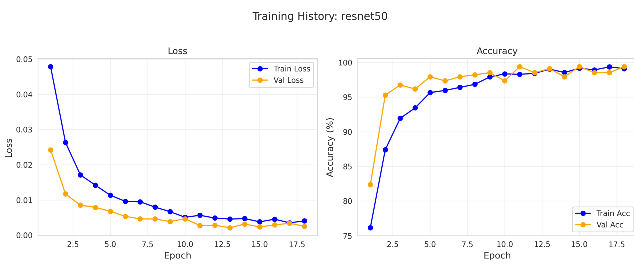
(k) ResNet50 (Learnable adjacency)

(l) ResNet50 (No GCN)

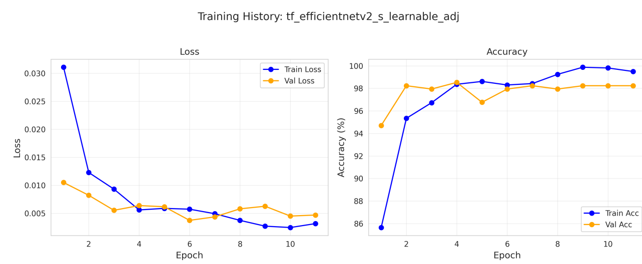


(m) ResNet50 (No MTL)

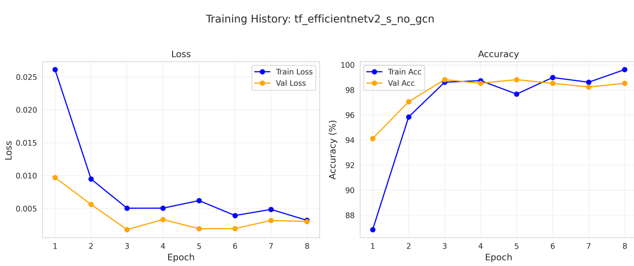
(n) ResNet50 (No SE)



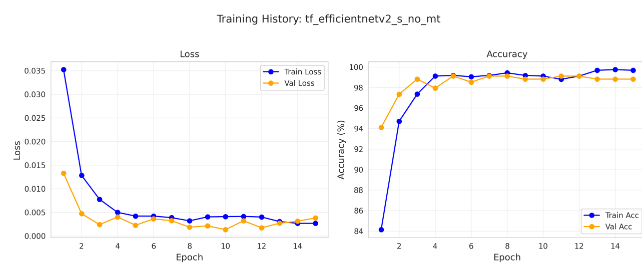
(o) ResNet50 (Baseline)



(p) EfficientNetV2-S (Learnable adjacency)



(q) EfficientNetV2-S (No GCN)



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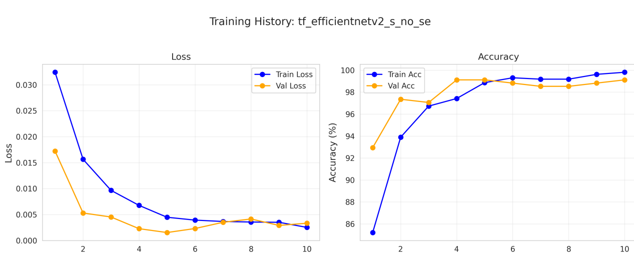

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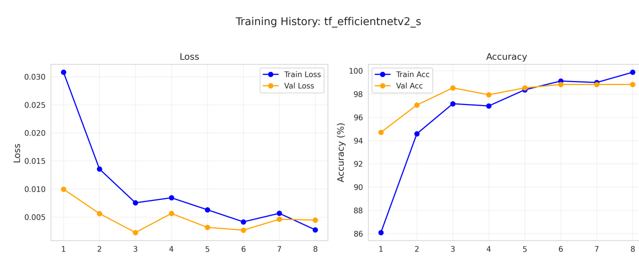
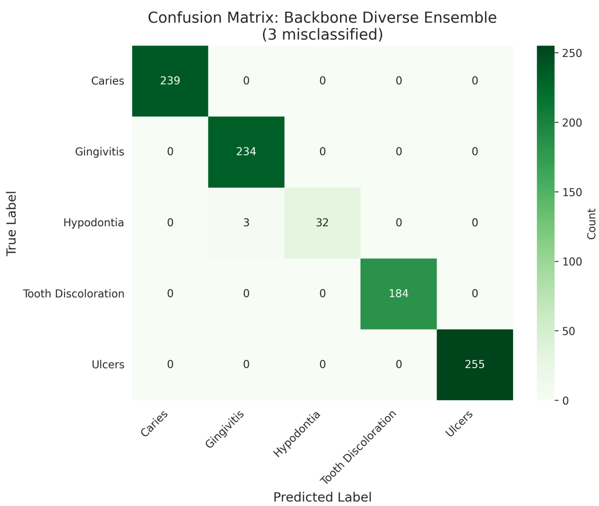


(t) EfficientNetV2-S (Baseline)

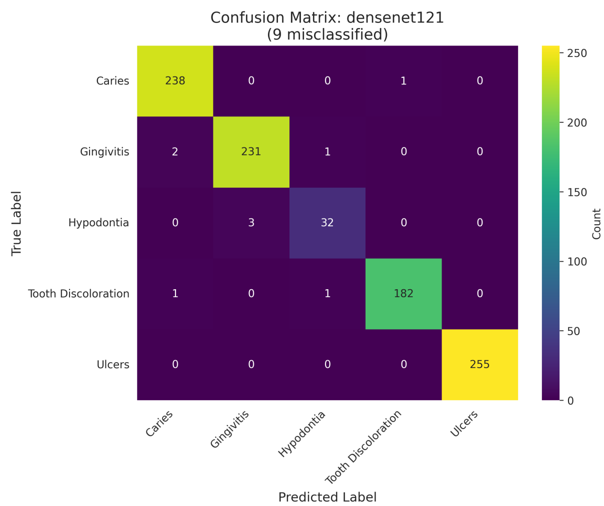
Supplementary Fig. S2. Learning curves for oral cancer dataset across model variants

1.2 Confusion Matrices

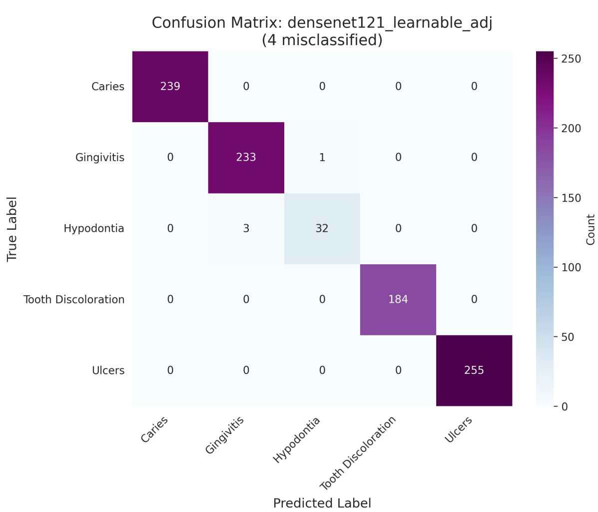
1.2.1 Dental Conditions



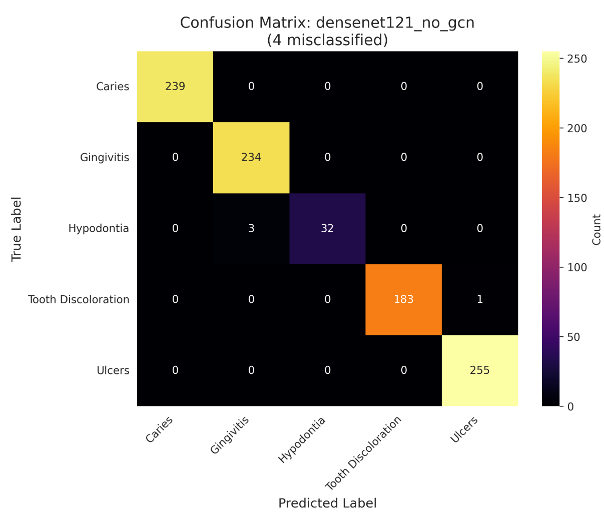
(a) Backbone diverse ensemble



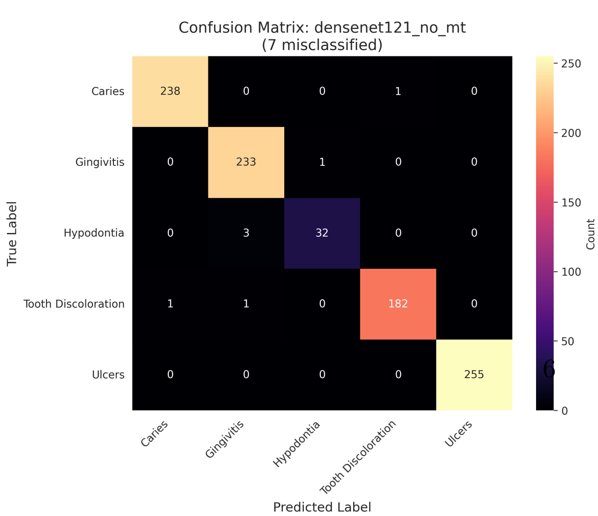
(b) DenseNet121 (Baseline)



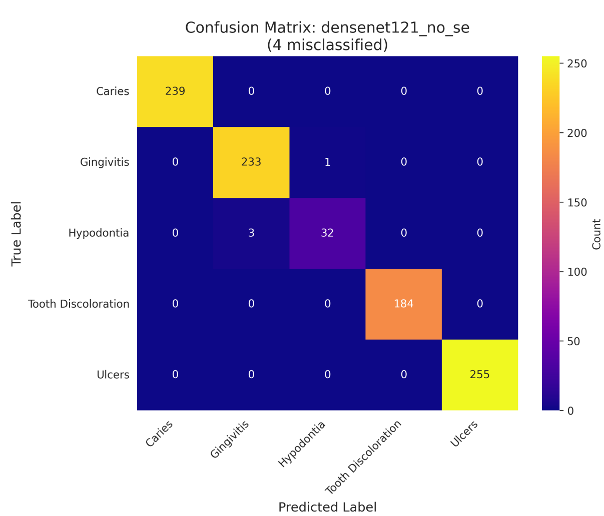
(c) DenseNet121 (Learnable adjacency)



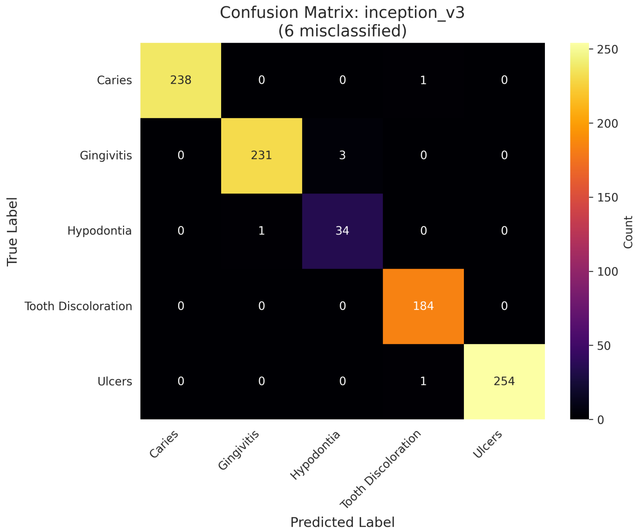
(d) DenseNet121 (No GCN)



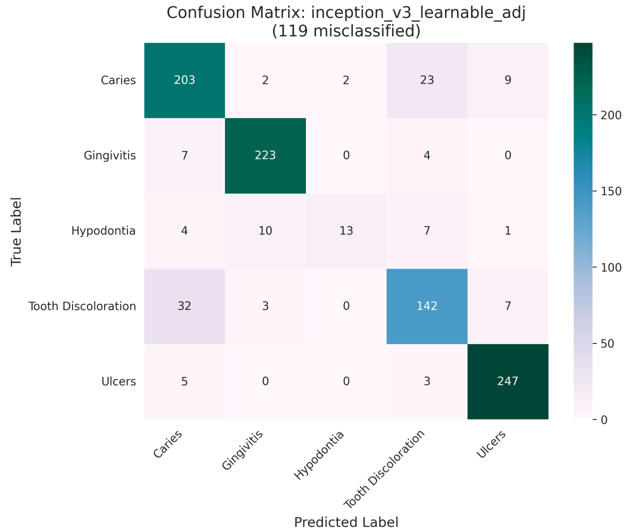
(e) DenseNet121 (No MTL)



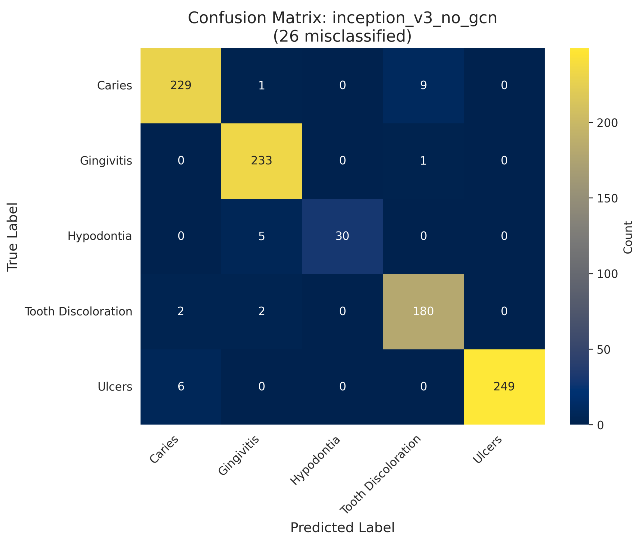
(f) DenseNet121 (No SE)



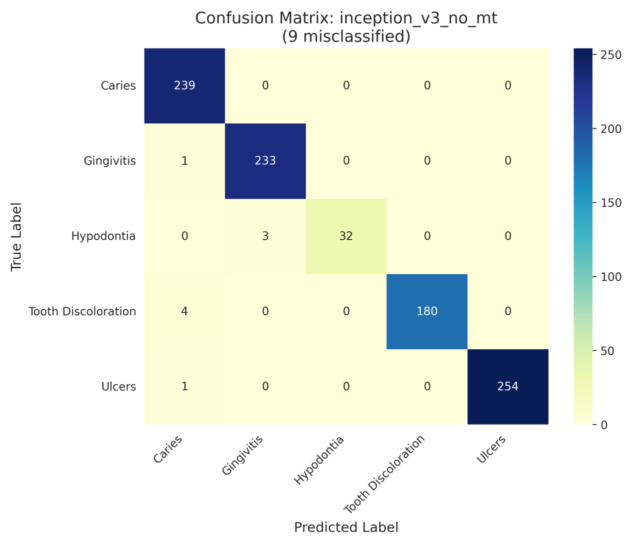
(g) Inception-V3 (Baseline)



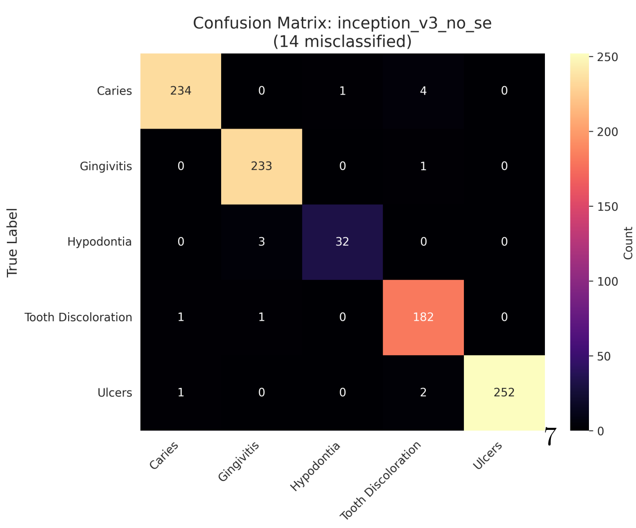
(h) Inception-V3 (Learnable adjacency)



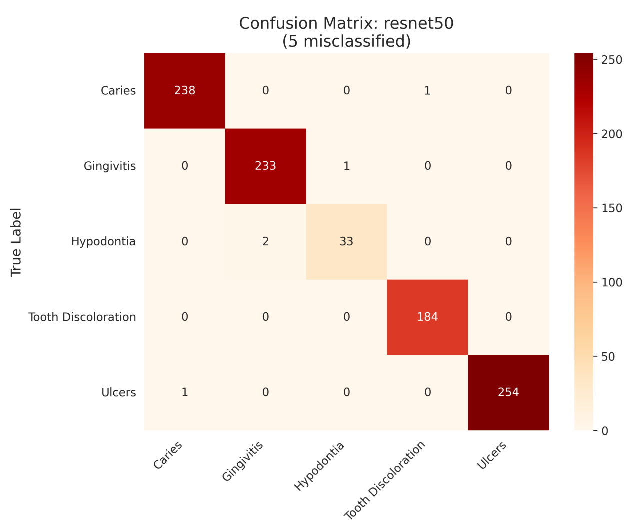
(i) Inception-V3 (No GCN)



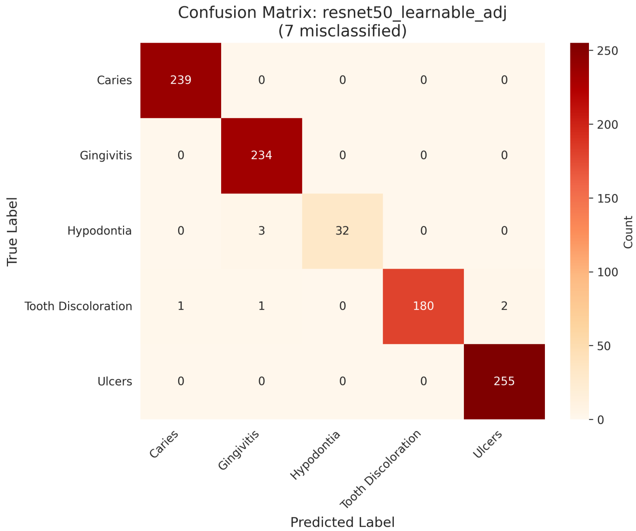
(j) Inception-V3 (No MTL)



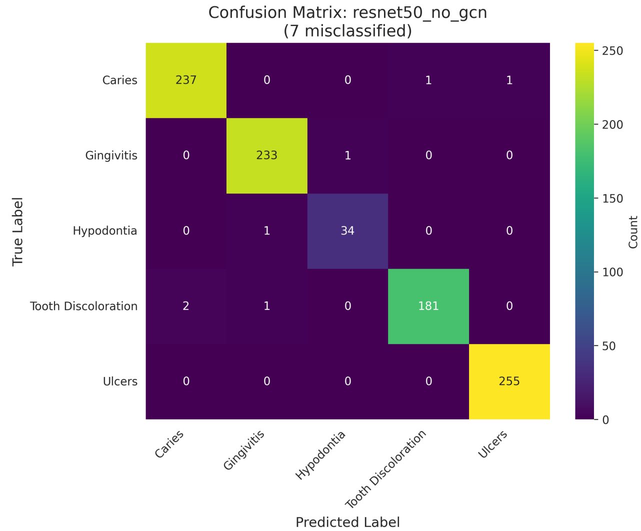
(k) Inception-V3 (No SE)



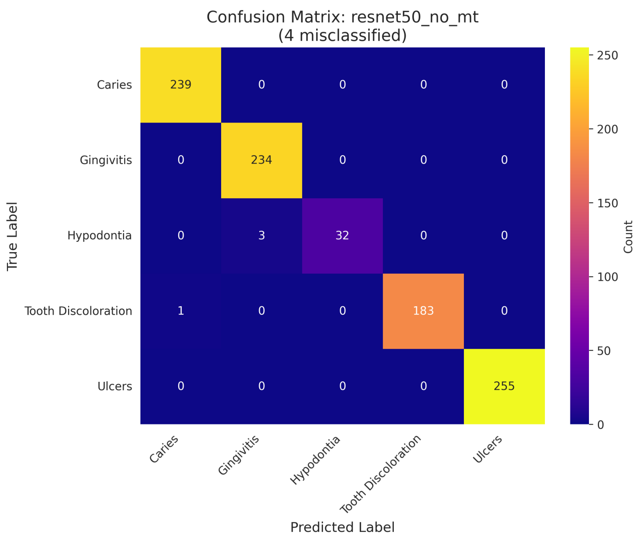
(l) ResNet50 (Baseline)



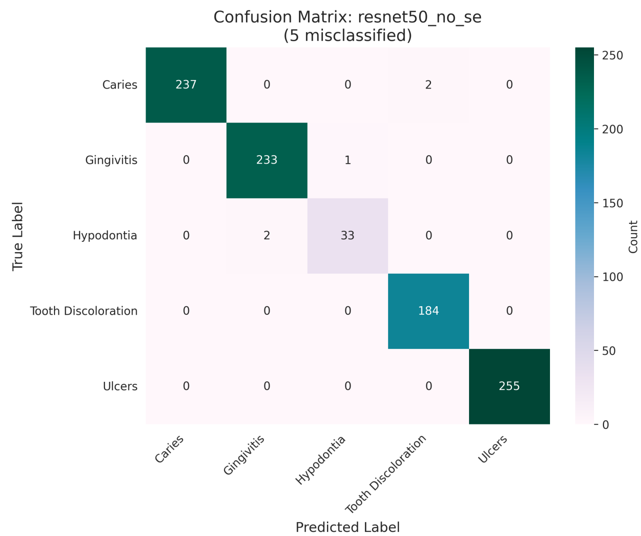
(m) ResNet50 (Learnable adjacency)



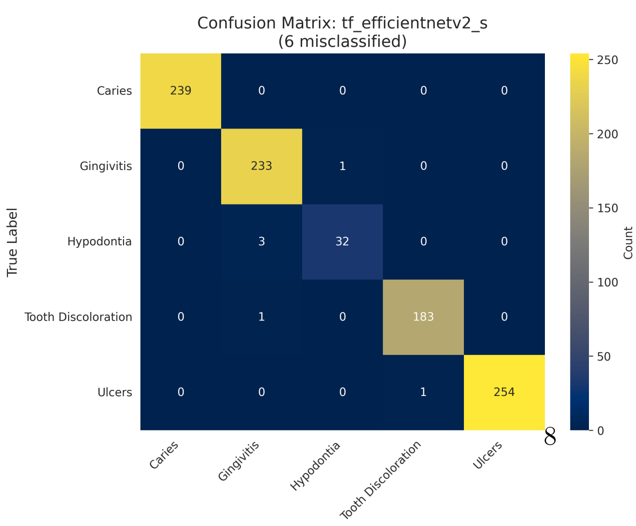
(n) ResNet50 (No GCN)



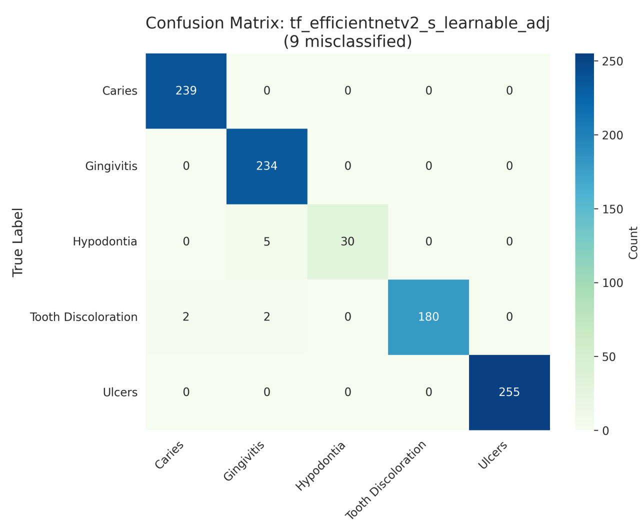
(o) ResNet50 (No MTL)



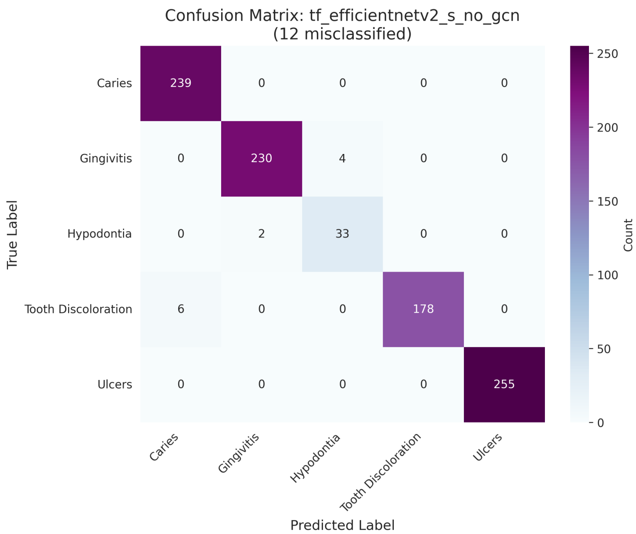
(p) ResNet50 (No SE)



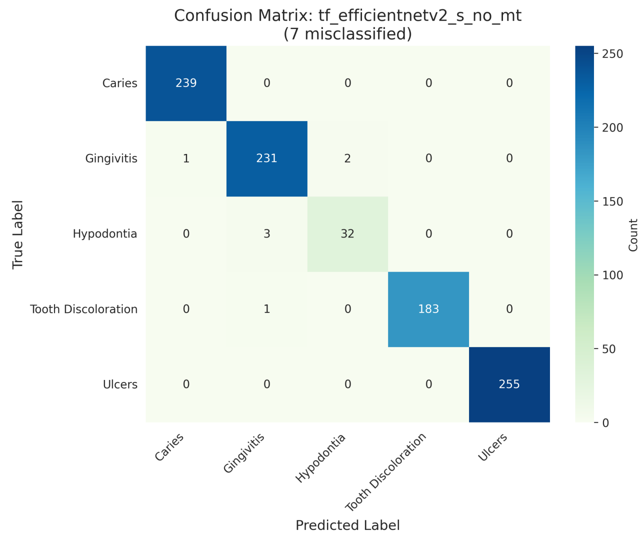
(q) EfficientNetV2-S (Baseline)



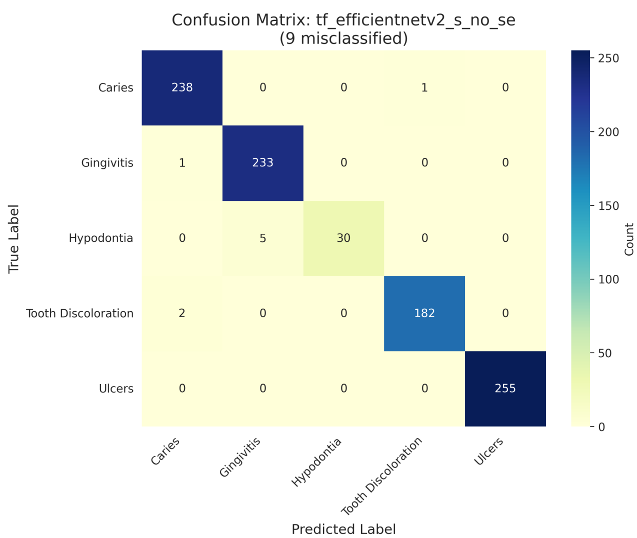
(r) EfficientNetV2-S (Learnable adjacency)



(s) EfficientNetV2-S (No GCN)



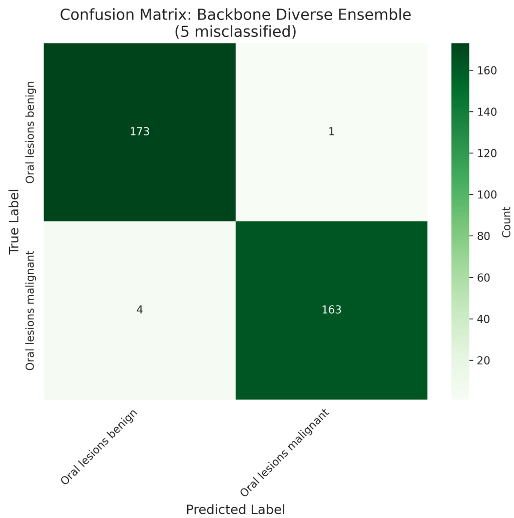
(t) EfficientNetV2-S (No MTL)



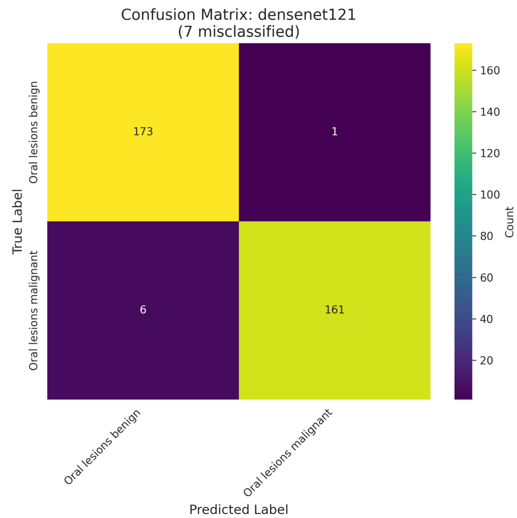
(u) EfficientNetV2-S (No SE)

Supplementary Fig. S3. Confusion matrices related to dental dataset with respect to different models

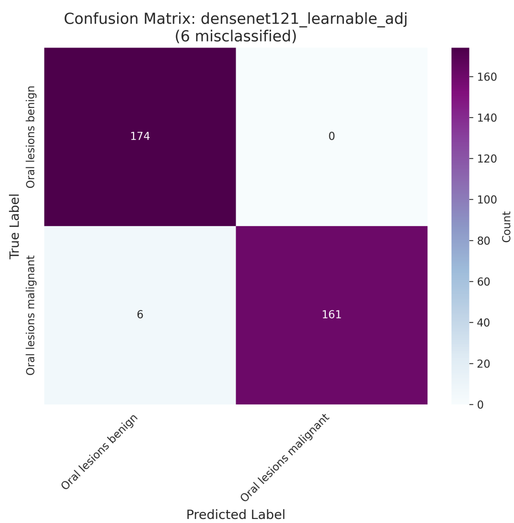
1.2.2 Oral Cancer



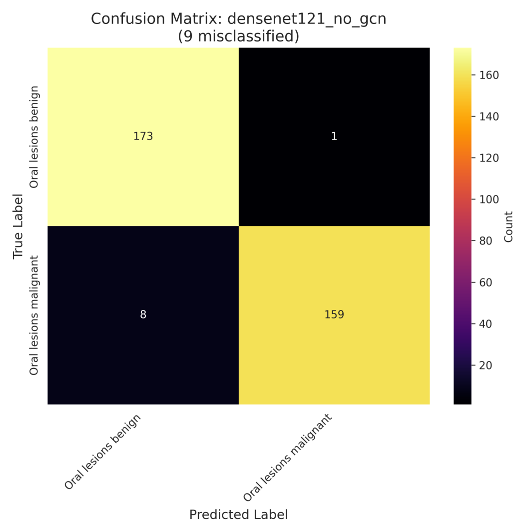
(a) Backbone diverse ensemble



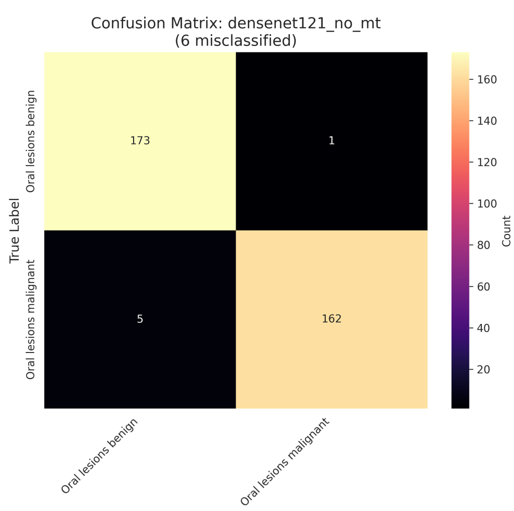
(b) DenseNet121 (Baseline)



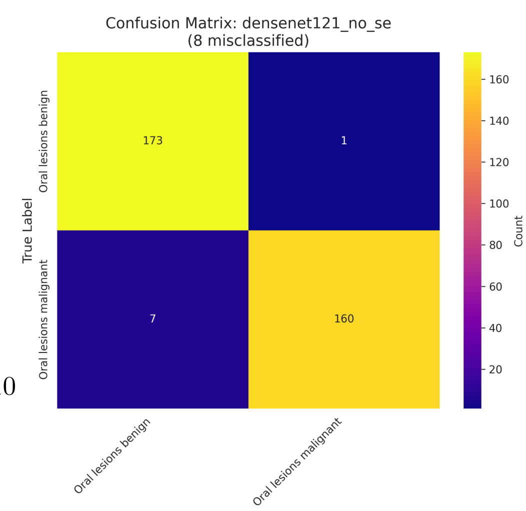
(c) DenseNet121 (Learnable adjacency)



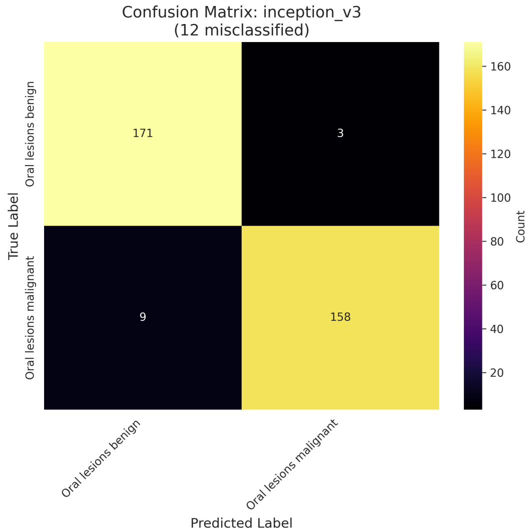
(d) DenseNet121 (No GCN)



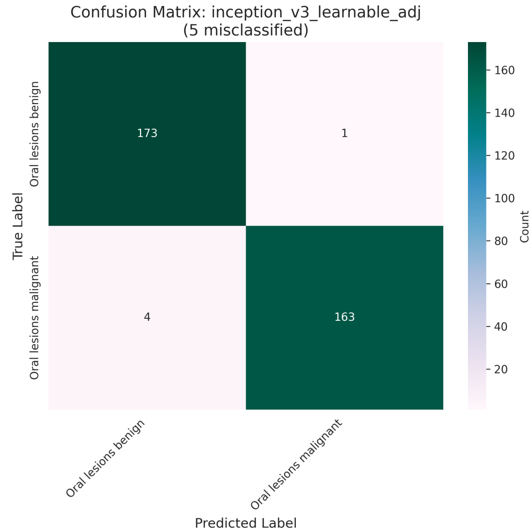
(e) DenseNet121 (No MTL)



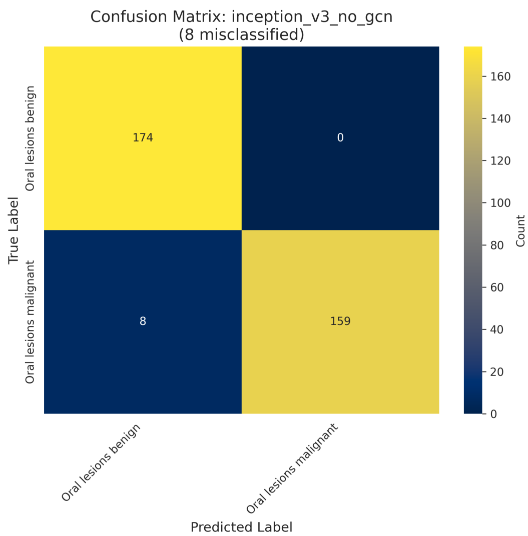
(f) DenseNet121 (No SE)



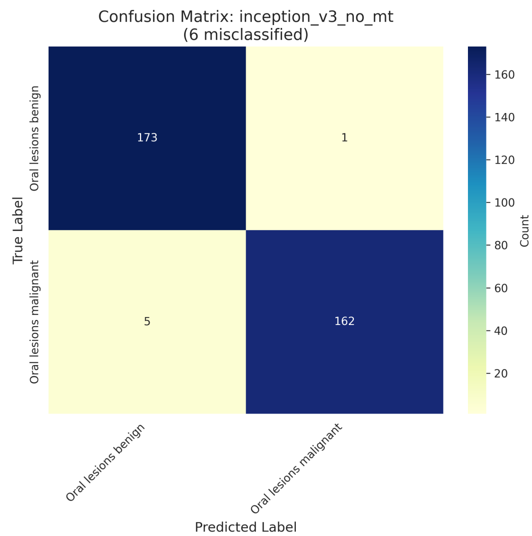
(g) Inception-V3 (Baseline)



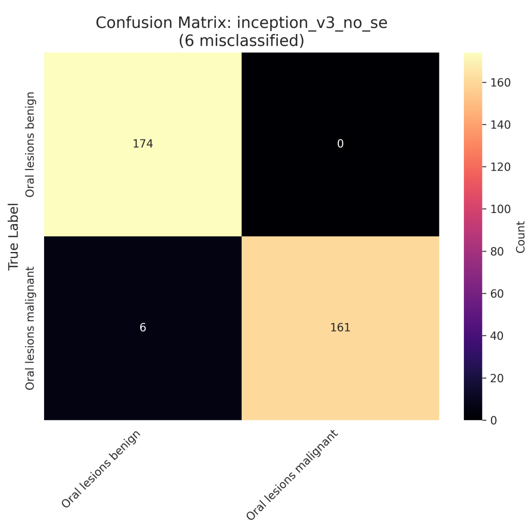
(h) Inception-V3 (Learnable adjacency)



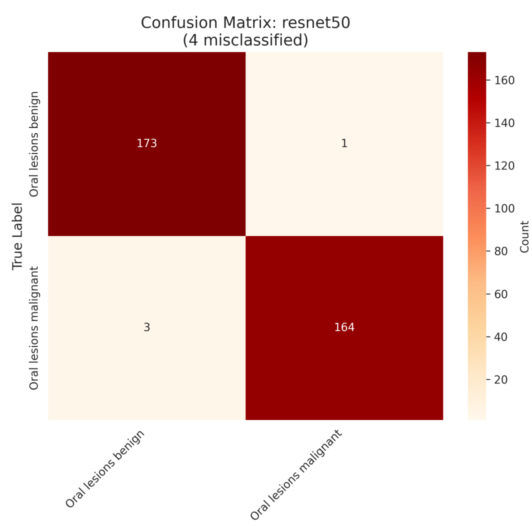
(i) Inception-V3 (No GCN)



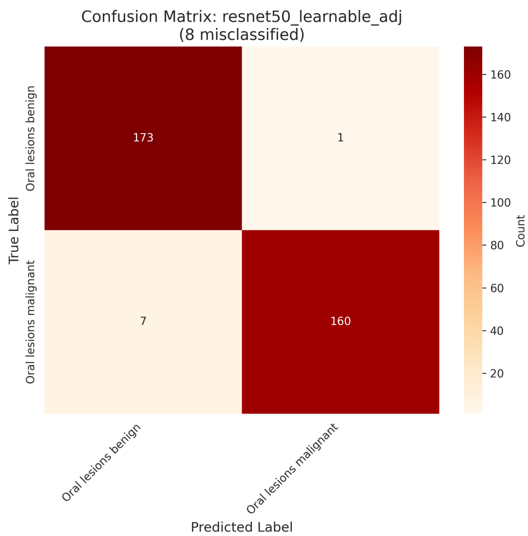
(j) Inception-V3 (No MTL)



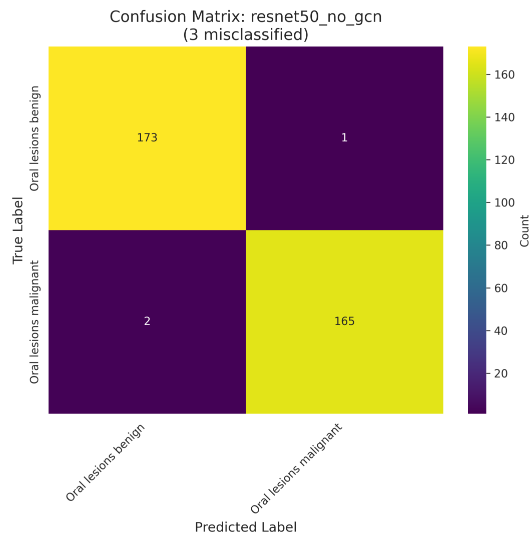
(k) Inception-V3 (No SE)



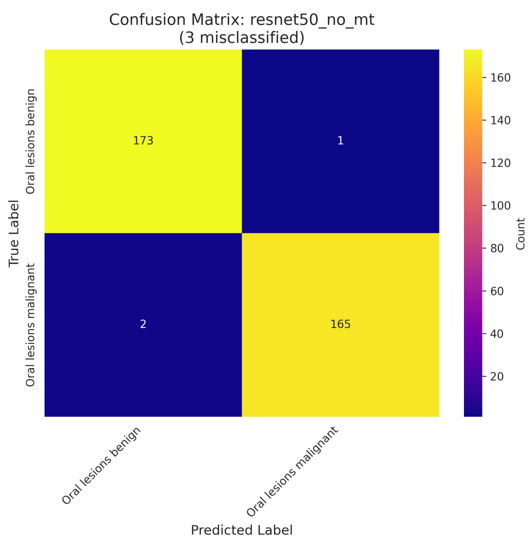
(l) ResNet50 (Baseline)



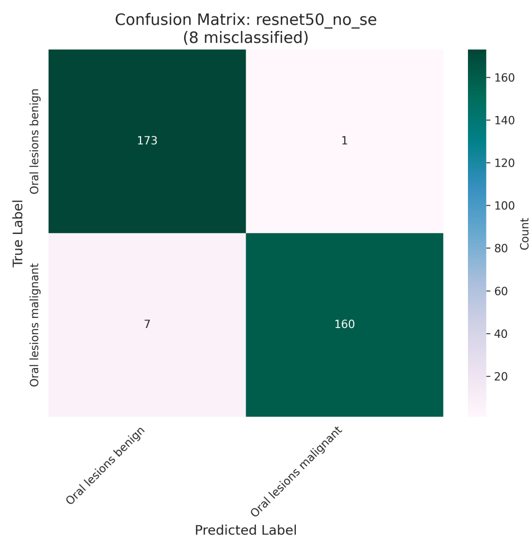
(m) ResNet50 (Learnable adjacency)



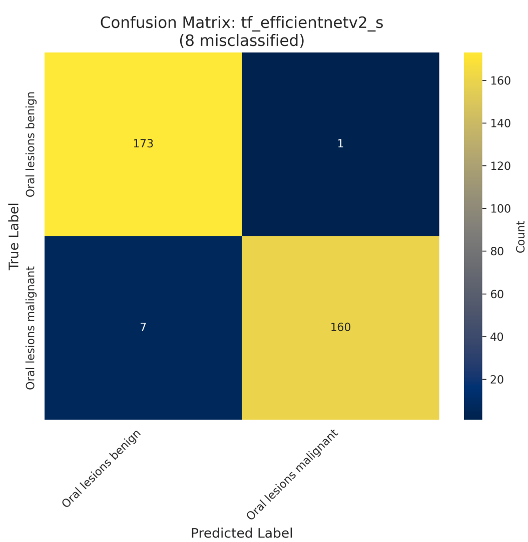
(n) ResNet50 (No GCN)



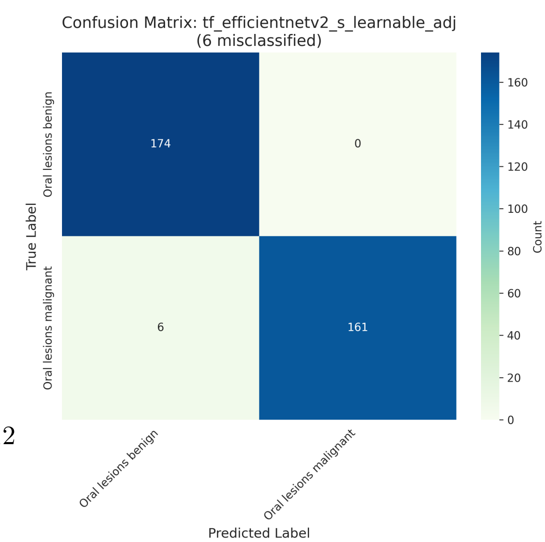
(o) ResNet50 (No MTL)



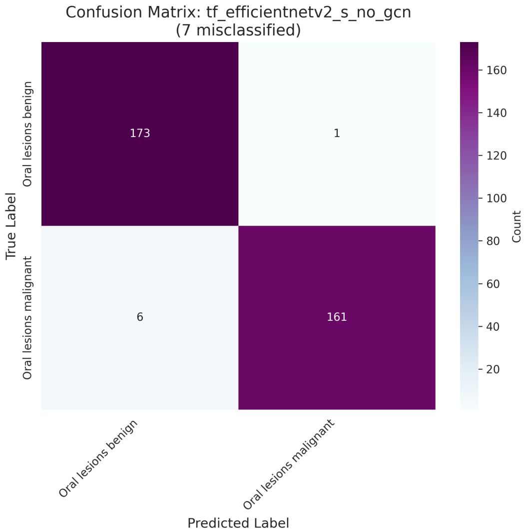
(p) ResNet50 (No SE)



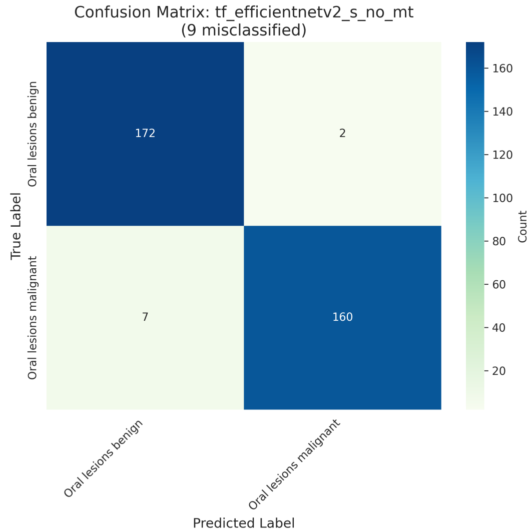
(q) EfficientNetV2-S (Baseline)



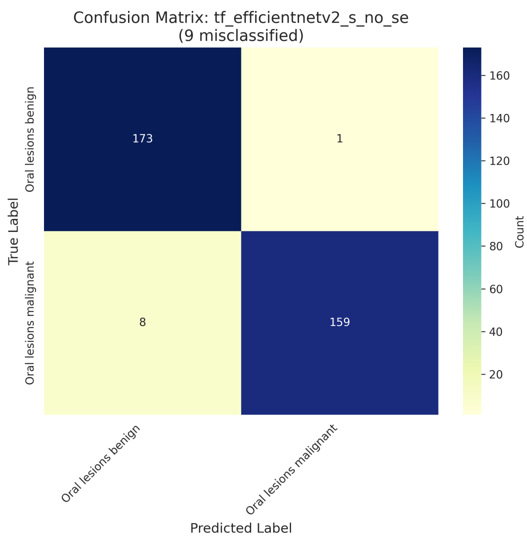
(r) EfficientNetV2-S (Learnable adjacency)



(s) EfficientNetV2-S (No GCN)



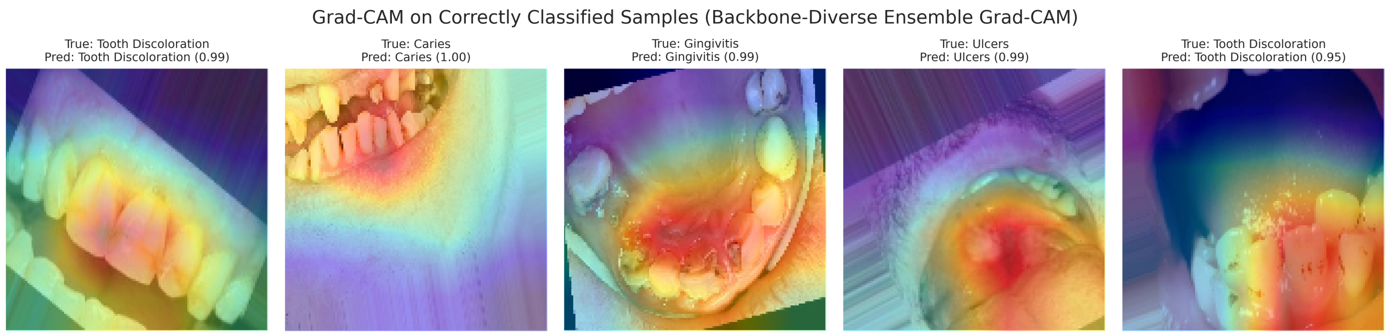
(t) EfficientNetV2-S (No MT)



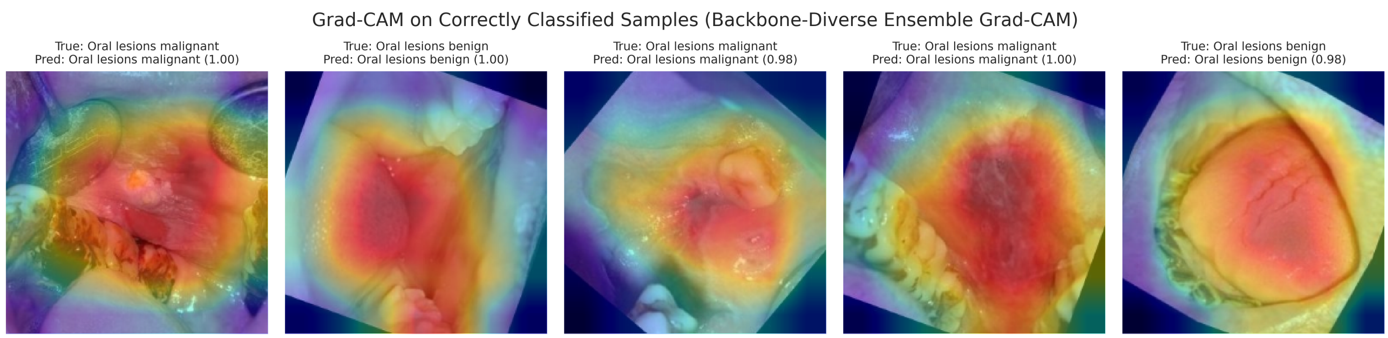
(u) EfficientNetV2-S (No SE)

Supplementary Fig. S4. Confusion matrices related to oral dataset with respect to different models

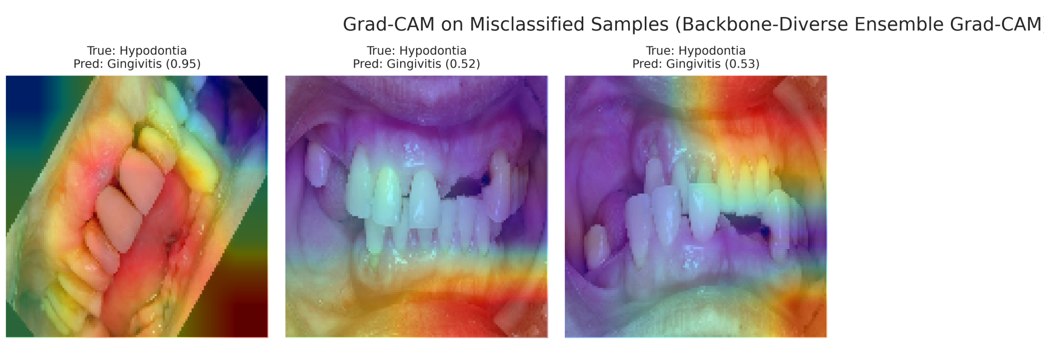
2 Supplementary Classification Examples



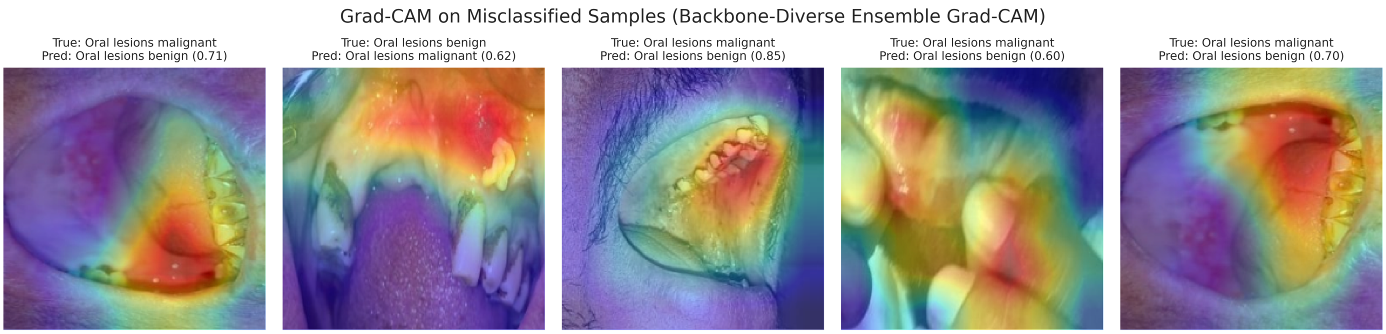
Supplementary Fig. S5. Grad-CAM on correctly classified samples of dental dataset



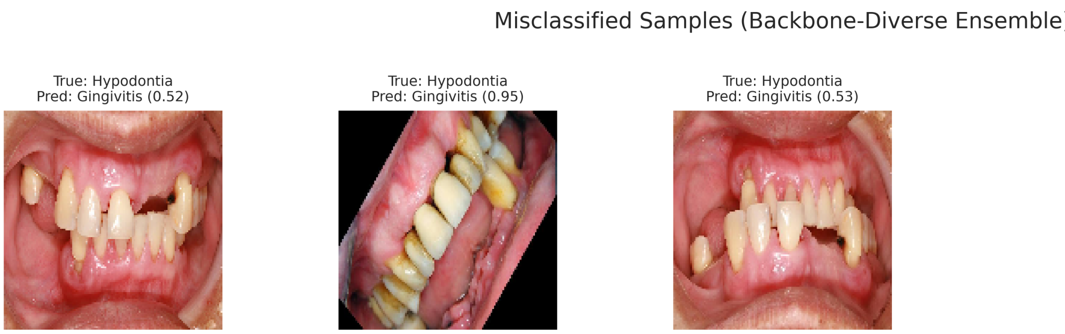
Supplementary Fig. S6. Grad-CAM on correctly classified samples of oral dataset



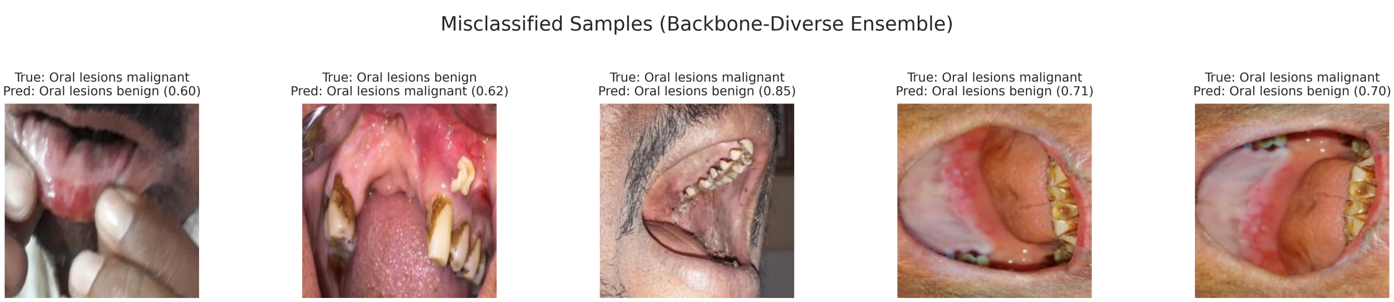
Supplementary Fig. S7. Grad-CAM on misclassified samples of dental dataset



Supplementary Fig. S8. Grad-CAM on misclassified samples of oral dataset



Supplementary Fig. S9. Misclassified samples of dental dataset



Supplementary Fig. S10. Misclassified samples of oral dataset

3 Supplementary Tables

3.1 Hyperparameter Configuration Table

Supplementary Table S1. Complete hyperparameter configuration

Component	Parameter	Value
Training	Initial learning rate	5×10^{-4}
	Weight decay	10^{-4}
	Batch size	32
	Maximum epochs	50
	Early stopping patience	5
Optimizer	AdamW β_1	0.9
	AdamW β_2	0.999
	Minimum learning rate	10^{-6}
	Cosine annealing T	50
Focal loss	Alpha (α)	0.25
	Gamma (γ)	2.0
	Label smoothing (ε)	0.1
SE block	Reduction ratio (r)	16
	Activations	ReLU + Sigmoid
Multi-Task	Hidden dimension	512
	Dropout Rate	0.6
Data augmentation	Input resolution	128×128
	Horizontal/Vertical flip probability	0.5
	Rotation limit	$\pm 20°$ (p=0.7)
	Color Jitter parameters	0.1 (p=0.5)
	Coarse dropout max holes	8 (p=0.5)
	CLAHE clip limit	2.0 (p=0.3)
	Elastic transform	$\alpha = 1$, $\sigma = 50$ (p=0.3)
TTA	Number of augmentations (K)	5
	TTA transformations	Flips (p=0.5), rotation $\pm 15°$ (p=0.5)
GCN	Number of classes (C)	5
	Adjacency epsilon (ε)	10^{-9}

Supplementary Table S2. Model weights assigned in the backbone-diverse ensemble for dental dataset

Model	Weight
densenet121__no__gcn	0.2010
tf__efficientnetv2__s__no__mt	0.2003
resnet50__no__gcn	0.2004
inception__v3__no__mt	0.1981
tf__efficientnetv2__s__learnable__adj	0.2003

Supplementary Table S3. False-Negative Rate (FNR) per class on test set

Class	FNR
Caries	0.0000
Gingivitis	0.0000
Hypodontia	0.0857
Tooth Discoloration	0.0000
Ulcers	0.0000

Supplementary Table S4. Detailed classification report for backbone-diverse ensemble on test set

Class	Precision	Recall	F1-Score	Support
Caries	1.0000	1.0000	1.0000	239
Gingivitis	0.9873	1.0000	0.9936	234
Hypodontia	1.0000	0.9143	0.9552	35
Tooth Discoloration	1.0000	1.0000	1.0000	184
Ulcers	1.0000	1.0000	1.0000	255
Accuracy			0.9968	947
Macro Avg	0.9975	0.9829	0.9898	947
Weighted Avg	0.9969	0.9968	0.9968	947

Supplementary Table S5. Model weights assigned in the backbone-diverse ensemble for oral cancer classification

Model	Weight
densenet121_learnable_adj	0.2012
tf_efficientnetv2_s_no_se	0.2000
resnet50	0.2000
inception_v3_learnable_adj	0.2012
tf_efficientnetv2_s_learnable_adj	0.1976

Supplementary Table S6. False-Negative Rate (FNR) per class on oral cancer test set

Class	FNR
Oral lesions benign	0.0057
Oral lesions malignant	0.0240

Supplementary Table S7. Detailed classification report for backbone-diverse ensemble on oral cancer test set

Class	Precision	Recall	F1-Score	Support
Oral lesions benign	0.9774	0.9943	0.9858	174
Oral lesions malignant	0.9939	0.9760	0.9849	167
Accuracy			0.9853	341
Macro Avg	0.9857	0.9852	0.9853	341
Weighted Avg	0.9855	0.9853	0.9853	341